

BRANDON REISS

Summit, NJ (NYC metro) | brandonreiss.com | LinkedIn

SUMMARY

Applied AI systems architect and former founder specializing in production ML platforms that handle real-world scale, latency, and cost constraints. Proven track record of operating autonomously to translate ambiguous R&D bets into durable production systems, architecting high-throughput unstructured data intelligence layers, and advising executive leadership on company-wide AI strategy.

EXPERIENCE

HubSpot | Remote

Jan 2025 – Present

Staff Software Engineer

- Architect and lead engineering for HubSpot's unstructured data signal extraction and intelligence layer, serving millions of monthly in-app CRM experiences (deal execution, meeting prep, account review).
- Integrated unstructured context into flagship AI assistant, setting evaluation standards from experimentation to release; achieved >20% daily traffic penetration with measurable lifts in CSAT and model quality benchmarks.
- Engineered unstructured content normalization and noise-filtering pipelines, significantly cutting inference token waste, reducing hallucinations, and establishing sustainable unit economics for LLM features.
- Advise cross-organizational teams on company-wide AI platform standards, focusing on model evaluation methodology, inference cost optimization, and low-latency architectural reliability.

Frame AI | New York, NY

Feb 2016 – Jan 2025 (*Acquired by HubSpot*)

Co-Founder / Principal Engineer / Director

- Co-founded and served as principal architect from inception through acquisition, scaling an applied AI platform that transformed unstructured conversational data into proactive, actionable business signals.
- Architected multi-stage ingestion, extraction, and synthesis pipelines processing continuous streams of customer interactions under demanding end-to-end SLAs.
- Implemented task prioritization and deduplication strategies across heterogeneous computational workloads and multi-tenant fault isolation across service fleet.
- Synthesized classical NLP and modern ML into configurable, cost-efficient production pipelines—unifying rules-based filtering, dependency parsing, sentence encoders, and LLMs into cohesive intelligence systems.
- Engineered storage backends combining relational databases (PostgreSQL) for transactional state with columnar and distributed object/NoSQL storage (ClickHouse) for high-volume analytical workloads.
- As Head of Security, established comprehensive security, compliance, and risk management programs; achieved and maintained SOC 2 Type II certification for 5 consecutive years with zero notable exceptions, delivered HIPAA compliance, and cleared Fortune 100 security audits.
- Partnered with research scientists, platform engineers, and enterprise partner teams to deploy resilient conversational intelligence solutions integrated with live products (e.g., Twilio Flex).

Knewton | New York, NY

Sep 2012 – Dec 2015

Software Engineering Manager / Senior Engineer

- Selected by senior leadership to lead premier engineering talent, bridging core applied R&D with production engineering across two teams in the Adaptive Learning division.
- Directed the architecture and operational delivery of statistical inference engines, large-scale recommendation systems, and automated pedagogy QA evaluation tools.
- Built core statistical modeling pipelines and distributed infrastructure powering the adaptive learning platform; measured statistical model drift and performance.

- Developed low-latency computer vision algorithms and GPU compute kernels for markerless, real-time 3D human motion tracking and clinical gait analysis.
- Built camera synchronization and distributed acquisition software for a 40-camera research array and an SLR-based 3D surface scanner.
- Designed and produced embedded hardware and firmware, including custom microcontroller PCBs mounted with rings of high-power near-IR LEDs and camera shutter sync routines.

TECHNICAL EXPERTISE & CORE COMPETENCIES

Agentic AI & Compound Systems → Token Budgeting & Context Window Management, Tool & Skill Architectures, Agent Task Performance Evaluation & Benchmarking, Task Aligned Data Retrieval, RAG.

Applied AI / ML & Statistical Inference → Recommender & Ranking Systems, Multi-Class Text Classification, Generalized Linear Models (GLMs), Signal Extraction & Feature Engineering, Sentence Encoders, Clustering, Syntactic/Dependency Parsing, Classical NLP & Rule-Based Heuristics.

Distributed Systems & Concurrency → Low-Latency Event-Driven Architectures, Multi-Tenant Resiliency (Blast-Radius Isolation, Anti-Correlated Failure Design), Priority-Tiered Processing, Fault-Tolerant Distributed Queues & Worker Architectures, Asynchronous / Parallel / Concurrent Execution Patterns, CPU & Memory Profiling.

Data Infrastructure & Storage → Relational Databases (PostgreSQL & InnoDB Internals, Concurrency, Lock Mechanics, Indexing), Columnar OLAP Datastores, Distributed Storage (HBase, S3).

Security, Compliance & Governance → SOC 2 Type II Program Architecture & Audit Defense, HIPAA Compliance, Enterprise Security Reviews, Risk Management & Data Governance, Application Security Testing.

Languages & Technical Tooling → Python, C++, Java, SQL, Bash / Linux CLI Tools; broad polyglot proficiency across modern software stacks; agentic coding harnesses & AI-assisted development workflows.

EARLY CAREER & APPLIED RESEARCH

Undergraduate Research Assistant

Rensselaer Polytechnic Institute (RPI)

Co-authored peer-reviewed research in Applied and Environmental Microbiology (April 2006, 72: 2460–2470) on isolating PCB-dechlorinating bacterial cultures using PCR, DNA sequence analysis, and gas chromatography.

Engineering Intern

BAE Systems

First high-school intern admitted in company history; designed a \$300k RF-absorbent antenna test facility and conducted experimental classification of RF signal polarization using neural networks.

EDUCATION

M.S. in Computer Science

New York University, Courant Institute of Mathematical Sciences

Recipient, Courant Hedwig Kurzbart Master's Fellowship (awarded annually for academic excellence)

B.A. in Computer Science and Mathematics

New York University, College of Arts and Science

Minor in Metropolitan Studies